

IB Portfolio, Explained

A plain-English guide to what you built, what the numbers mean, and how to answer when someone asks

This guide has one job: make sure that when an interviewer points at any cell in any of your models, you can say what it is, where it came from, and why you chose it. A portfolio you cannot defend is worse than no portfolio, because it turns a neutral first impression into a bad one in about ninety seconds.

It is written in ordinary language. Where a piece of jargon matters, it is explained the first time it appears and then used normally, because you will need to use it normally too.

How to use this

Read Part 1 once to fix the vocabulary. Read the Part 2 section for a project the night before you discuss it. Part 3 is the question bank — practise saying those answers out loud, not just reading them. If you can answer Part 3 without looking, you own the portfolio.

What is in here

Part	What it covers
1	The vocabulary — twelve terms that carry almost everything
2	Project by project — what each one is, in one line and then properly
3	The questions you will actually be asked, with your answers
4	The three rules that make this portfolio different, and why they matter

Figures throughout are the ones actually in your models, as of the 12 September 2026 market date used across the repository. If you rebuild a model and a number moves, update it here too.

Part 1 · The vocabulary

Almost all of investment banking valuation runs on about twelve ideas. Everything else is detail.

1. Equity value and enterprise value

Equity value is what the shareholders own: share price times number of shares. It is what you would pay to buy all the shares.

Enterprise value is what the whole business is worth regardless of who financed it: equity value plus debt, minus cash. The logic is that if you buy a company you inherit its debts, so you must pay for those too — but you also get its cash, so that comes off.

The one-line version

Buying a house for 1 million with a 300,000 mortgage attached and 50,000 in cash in the kitchen drawer: equity value 1 million, enterprise value 1,250,000. You pay the seller, you take on the mortgage, you keep the cash.

This matters constantly. Profit measures before interest (like EBITDA) belong to everybody — lenders and shareholders — so they pair with enterprise value. Profit measures after interest (like net income) belong only to shareholders, so they pair with equity value. Mixing them is the most common junior error.

2. EBITDA

Earnings before interest, tax, depreciation and amortisation. A rough measure of the cash a business generates from operations before financing and accounting choices get involved. Bankers like it because it lets you compare two companies with different debt loads and different depreciation policies.

It is not cash flow. It ignores the money spent on equipment and the money tied up in unpaid customer invoices. Anyone who tells you EBITDA is cash flow has not run a business.

3. Multiples

A multiple is a price expressed in years of profit. If a company is worth 10 billion and makes 1 billion of EBITDA, it trades at 10 times EBITDA. Saying “10 times” is quicker than saying the price, and it lets you compare a large company with a small one.

The two you will use most: **EV / EBITDA** and **P / E** (share price divided by earnings per share). The first values the whole business, the second values the shares only.

4. Discounted cash flow, or DCF

The idea: a business is worth the cash it will produce in future, but future cash is worth less than cash today, so you shrink each future year by a discount rate before adding them up.

Money today is worth more for two reasons: you could invest it, and the future is uncertain. The discount rate carries both.

5. WACC — the discount rate

Weighted average cost of capital: the blended return the company's investors require. If a business is funded half by shareholders wanting 12% and half by lenders charging 6%, its WACC is roughly 9%.

The shareholder half comes from CAPM: risk-free rate, plus beta times the equity risk premium.

- **Risk-free rate** — what the government pays to borrow for ten years. Your safest alternative.
- **Equity risk premium** — the extra return investors demand for owning shares instead of government bonds. Around 4.5% in developed markets; more where the country is riskier.
- **Beta** — how much this share moves when the whole market moves. Beta of 1 means it moves with the market; 0.9 means it moves a bit less; 1.5 means it swings harder.

6. Terminal value

A DCF forecasts five years in detail, then has to account for every year after that. The lump sum representing all of it is the terminal value. It is usually 60–80% of the answer, which is worth admitting rather than hiding: most of a DCF is an assumption about a distant year, not a forecast.

7. Free cash flow

The cash left after the business has paid its operating costs, its taxes, bought the equipment it needs, and funded the extra working capital that growth requires. It is what a DCF discounts, and what an LBO uses to repay debt.

8. Working capital

Money stuck in the business day to day — mostly customers who have not paid yet, less suppliers you have not paid yet. When a company grows, this number grows, and that growth consumes cash. That is why a profitable, fast-growing company can still run out of money.

9. Comparable companies and precedent transactions

Trading comparables ask: what is the stock market paying today for similar businesses? **Precedent transactions** ask: what have buyers actually paid to acquire similar businesses?

Precedents are almost always higher, because a buyer taking control pays a premium for it. The gap between the two is the control premium.

10. Accretion and dilution

In a takeover: does earnings per share go up or down for the buyer's existing shareholders? If up, the deal is accretive; if down, dilutive.

The mechanism is simple. The buyer gains the target's profits but usually issues new shares to pay for it. If profits rise by more than the share count, EPS rises. If not, it falls.

11. Tangible book value, and why banks are different

Book value is what the accounts say the shareholders own. **Tangible** book value strips out goodwill and other intangibles — things you cannot sell separately.

For banks this is the number that matters, because regulators require a bank to hold real capital against its loans, and goodwill does not count. A bank deal that destroys tangible book destroys the bank's capacity to lend, however good the earnings look.

12. IRR and MOIC

MOIC — multiple on invested capital. Put in 100, get back 200, that is 2.0 times.

IRR — internal rate of return: the same result expressed as an annual percentage. 2.0 times over five years is about 15% a year. Private equity funds typically target 20%+.

Why both

MOIC ignores time: doubling your money in two years and in ten years are both 2.0 times. IRR fixes that but flatters short deals. Bankers quote both because neither alone is honest.

Part 2 · Project by project

Seven projects are built. Each section gives the one-line version, what it actually does, the numbers that came out, and the single thing that makes it worth showing to someone.

Project 1 · Apple — three-statement operating model

One line: a forecast of Apple's profit, balance sheet and cash, wired together so changing one assumption updates all three.

The three statements are the income statement (did we make money), the balance sheet (what do we own and owe) and the cash flow statement (where did the cash actually go). In a real model they are linked: profit flows into retained earnings on the balance sheet, and cash flow explains the change in the cash line.

The test of a three-statement model is that the balance sheet balances without you forcing it. If it does, your links are right. If you plug a number to make it balance, the model is decoration.

Yours has a scenario switch — one cell changes the whole model between base, upside and downside — and a revolver, which is an automatic overdraft that kicks in if the company would otherwise run out of cash.

Why it matters

This is the foundation model in banking. Everything else — DCF, LBO, merger — sits on top of one of these. Building one that ties is the basic proof that you can do the job.

Project 2 · Apple — discounted cash flow valuation

One line: what is Apple actually worth, based on the cash it will generate, rather than what the market says today.

Takes the forecast from Project 1, works out the free cash flow each year, and discounts it back at Apple's WACC. The WACC was built from scratch rather than assumed: ten peer companies, five years of weekly share price moves regressed against the market to get each one's beta, adjusted for how much debt each carries, then the median applied to Apple.

Item	Value	Plain English
Beta	1.12	Apple swings slightly harder than the market
Cost of equity	10.0%	What shareholders require to hold it
WACC	9.9%	Apple is almost all equity-funded, so the two are nearly the same

Why it matters

Terminal value is calculated two ways — by assuming perpetual growth, and by assuming a sale at an exit multiple — and each is checked against what the other implies. If assuming 3% growth forever implies selling at 25 times earnings, and you thought 12 times was right, one of your two assumptions is wrong. Most student DCFs never run that check.

Project 3 · TCS — trading comparables

One line: what the stock market pays for IT services companies today, and what that implies TCS is worth.

Eight peers across three tiers: Indian large caps, Indian mid caps, and global names like Accenture and Cognizant. For each, enterprise value is built up properly and four multiples calculated.

Two technical problems were solved here, and both are worth mentioning in an interview:

- **Currency.** Infosys reports its accounts in US dollars but its shares trade in rupees. Multiply those together carelessly and the multiple is wrong by a factor of 85. The model detects the reporting currency and normalises everything to dollars before any division happens.
- **Calendarisation.** TCS's year ends in March, Accenture's in August, Cognizant's in December. Comparing forward multiples across them compares different twelve-month periods. The model weights each peer's forecast years to line them all up with TCS's March year.

Why it matters

Anyone can pull multiples off a screen. Knowing that they are not comparable until you fix currency and fiscal year ends is the difference between a student and an analyst.

Project 4 · IT services — precedent transactions

One line: what buyers have actually paid for IT services businesses over ten years.

Fourteen deals from 2014 to 2024 — Capgemini buying IGATE, Atos buying Syntel, LTI merging with Mindtree, Cognizant buying Belcan and others — built from the original announcement press releases and filings rather than downloaded from a database.

Measure	25th percentile	Median	75th percentile
EV / revenue	1.8x	2.6x	3.4x
EV / EBITDA	13.6x	14.7x	16.9x
One-day premium	3.5%	4.6%	9.0%

Why it matters

The notes say plainly that part of the apparent control premium is not control at all — it is the cycle. Those deals were struck when the sector traded at 20–30 times earnings; it now trades at 10–15. Comparing deal multiples from one era with trading multiples from another overstates the premium. Spotting that is the kind of thing that gets remembered.

Project 5 · TCS — football field and valuation summary

One line: every method of valuing TCS on one chart, as ranges rather than single numbers.

A football field is the chart bankers put in front of clients: one horizontal bar per valuation method, each running from low to high. You read the overlaps. Where independent methods agree, you have something.

Method	Low	High	Weight
52-week trading range	1,977	3,350	0%
Analyst price targets (43)	1,775	3,900	10%
DCF	1,625	2,504	35%
Trading comps (three multiples)	1,242	2,689	45%
Precedent transactions	1,428	3,532	10%
Recommended range (weighted)	1,676	2,614	

All figures INR per share. The share price on the day was 2,201, which sits 2.6% above the weighted midpoint of 2,145.

Why it matters

Two things. First, the weights are inputs on the sheet, not hidden inside a formula — a client can argue with them, which is the conversation you want. Second, and better: the standard beta method from Project 2 was tried here and **rejected**. Regressing Indian shares against the S&P 500 produced betas of 0.06 to 0.33 with R-squared of 0.00 to 0.04 — meaning the relationship explains essentially nothing, because the two markets trade in different hours and different currencies. The regression was redone against the Indian market instead, giving 0.91 with R-squared 0.28. The rejected attempts are still on the sheet.

If you remember one thing for interviews

“R-squared” measures how much of one thing’s movement is explained by another, from 0 to 1. An R-squared of 0.01 means your beta is noise. Knowing to throw a result away is a stronger signal than producing one.

Project 6 · Emirates NBD / Mashreqbank — merger model

One line: should Emirates NBD buy Mashreqbank, and the answer is no.

A full accretion/dilution model with real purchase accounting. Bank deals are judged in a specific order, and the model follows it: tangible book dilution first, regulatory capital second, earnings third.

Test	Result	What it means
Price paid	2.16x tangible book	Emirates NBD’s own shares trade at 1.41x
EPS, year 3	(2.3%)	Still dilutive even at full synergies
Break-even synergies	44.6% of target costs	Against 25% assumed. Not achievable
Tangible book dilution	(16.7%)	Shareholders lose a sixth of their tangible book
Earnback	Never	Earnings never exceed standalone, so it is never recovered
Pro forma CET1	13.0%	Down 2 points; survives, but only just
Return on capital	8.6%	Against a 12% required return

Why it matters

An acquirer whose shares trade at 1.4 times tangible book, issuing those shares to buy assets at 2.2 times, hands value to the seller the moment the deal is announced. No amount of synergy phasing reverses it.

Rather than reporting “2.3% dilutive” and inviting an argument about whether synergies could be higher, the model solves for the level that would make it work: 44.6%. That closes the discussion. **A merger model that only ever says yes is not being used properly.**

CET1 is a bank’s core capital as a percentage of its risk-weighted loans. Regulators set a floor; here it is 11%. Goodwill is deducted in full, which is why an expensive bank acquisition eats capital.

Project 7 · Cognizant — leveraged buyout

One line: could a private equity firm take Cognizant private, and what is the most it could pay?

An LBO is buying a company mostly with borrowed money, using the company’s own cash flow to repay the debt, and selling it five years later. The return comes from three places: debt repaid, profit grown, and the multiple changing.

Item	Value
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Offer price (30% premium)	US\$81.36 per share
Entry multiple	9.58x EBITDA
Debt raised	5.25x EBITDA — US\$21.3bn
Sponsor equity cheque	US\$18.9bn
Exit at 9.00x in year 5	MOIC 2.00x, IRR 14.9%
Versus the 20% hurdle	Misses by 5.1 points
Maximum price at 20% IRR	US\$73.78 — an 18% premium

Why it matters

The exit multiple is set *below* the entry multiple deliberately. Assuming you sell for more than you paid is how an LBO model is made to produce whatever answer is wanted, and one of the sixteen checks fails the model if anyone raises it above entry.

The genuinely useful output is not the IRR. It is the maximum price — US\$73.78 — because that is the floor under the share price in any sale process. And it sits 18% above where Cognizant actually trades, which says the market is pricing it below what a leveraged buyer could justify.

Part 3 · The questions you will be asked

These are the questions an interviewer reaches for when they see this repository. Practise them out loud.

General

Q. Walk me through a DCF.

Forecast free cash flow for five years. Discount each year back at WACC. Calculate a terminal value for everything after year five, either by assuming perpetual growth or by applying an exit multiple, and discount that back too. Add them up to get enterprise value. Add net cash or subtract net debt to get equity value, then divide by shares for a per-share number.

Q. Why would you use EBITDA rather than net income?

EBITDA sits above interest and tax, so it is not affected by how a company is financed or where it is domiciled. That makes two companies with different debt loads comparable. The cost is that it ignores capital spending and working capital, so it is not cash flow.

Q. Which is higher, precedent transaction multiples or trading multiples, and why?

Precedents, normally, because an acquirer taking control pays a premium for synergies and for the right to run the business. In my Project 4 the gap is 2.6 times revenue against 1.8 times. But I would add that part of that gap is the cycle, not control — those deals were struck when the sector traded much higher, and comparing across eras overstates the premium.

Q. A company has negative free cash flow but is profitable. How?

Usually working capital. If it is growing fast, customers owe it more each month and inventory rises, and that swallows cash before it ever reaches the bank. Heavy capital spending does the same.

Project 2 — the Apple DCF

Q. Where did your WACC come from?

Bottom-up. I took ten peers, regressed five years of weekly returns against the S&P 500 to get each one's beta, unlevered each at its own net debt to equity, took the median, then relevered at Apple's capital structure — which is zero, because Apple is net cash. Then Blume-adjusted, which is two-thirds raw beta plus one third, the way banks do it. That gave 1.12, a cost of equity of 10.0% on a 4.95% ten-year Treasury and a 4.5% equity risk premium, and a WACC of 9.9%.

Q. Why is your WACC almost the same as your cost of equity?

Because equity is about 98% of Apple's capital. There is barely any debt to weight in.

Project 5 — the TCS football field

Q. Why is your TCS discount rate 13.3% when Apple's was 9.9%?

Different country and different currency. TCS's cash flows are in rupees, so I discount at a rupee cost of capital: a 7.02% Indian ten-year government bond rather than a 4.95% US Treasury, and an equity risk premium of 6.7% rather than 4.5%, because that adds India country risk. Discounting rupee cash flows at a dollar rate would bury an exchange-rate forecast inside the valuation.

Q. Why 5.5% terminal growth?

Roughly Indian long-run inflation plus a little real growth, and comfortably below India's nominal GDP growth — nothing can grow faster than its economy forever. I also cross-check it: 5.5% implies a 9.1 times exit multiple, and the 11 times exit multiple implies 6.8% growth. The two are close enough that neither contradicts the other.

Q. Your DCF says 1,953 and the stock is 2,201. Is TCS a sell?

I would not put it that strongly. My weighted range is 1,676 to 2,614 and the price sits inside it, about 2.6% above the midpoint. The honest reading is that the market is paying a modest premium to both my intrinsic value and the peer group. Whether that is wrong depends on one specific question — whether the sector's AI de-rating is permanent — and my model does not answer that.

Q. Seventy per cent of your DCF is terminal value. Isn't that a problem?

It is a limitation and I say so on the sheet. It is normal for an asset-light business with low capital intensity, but it does mean most of the answer is an assumption about 2031. That is exactly why I show the DCF as a sensitivity range on the football field rather than as a single point.

Project 6 — the bank merger**Q. Why do banks care about tangible book value more than EPS?**

Because regulatory capital is built from tangible equity. Goodwill is deducted in full from CET1, so paying a large premium for a bank converts real capital into an intangible that cannot support any lending. A deal can look accretive to earnings and still leave the bank less able to write loans.

Q. Your model says the deal fails. Why build it?

Because that is the answer. Emirates NBD trades at 1.4 times tangible book and would be paying 2.2 times. Rather than stop at 'dilutive', I solved for the synergies needed to break even — 44.6% of Mashreq's cost base, roughly double what is assumed and above anything disclosed in comparable bank mergers. In a real process that number is what you take to the client.

Q. Why did you set revenue synergies to zero?

Because they are the standard way this model gets made to work — a cross-sell assumption big enough to close the gap with no mechanism behind it. I left the line in at zero so the choice is visible. It would not rescue this deal anyway; the shortfall is about AED 700m after tax.

Q. Why is the consideration 70% stock?

Capital. At that mix pro forma CET1 lands at 13.0%, about two points above the minimum. A cash-heavy structure would not clear the floor, because cash paid away does not come back as capital.

Project 7 — the LBO**Q. What drives returns in an LBO?**

Three things: repaying debt with the company's own cash flow, growing EBITDA, and any change in the multiple between buying and selling. In mine the bridge is roughly US\$11.4bn from debt paydown and US\$11.2bn from EBITDA growth, less US\$2.4bn lost to multiple contraction.

Q. Why did you assume a lower exit multiple than entry?

Because assuming multiple expansion is how an LBO model is made to say yes. If the return only works when you sell higher than you bought, you have not analysed anything. I also put a check in the model that fails if anyone later raises the exit multiple above entry.

Q. Why does interest use the opening balance rather than the average?

Average balances make the model circular — interest drives cash flow, cash flow drives the sweep, the sweep drives interest. Resolving that needs iterative calculation enabled, and a workbook that only computes correctly with a setting switched on will break on someone else's machine. Using the opening balance slightly overstates interest, which is conservative.

Q. What is a cash sweep?

A term in the loan requiring spare cash to go toward repaying debt rather than to the owners. Mine applies 100% of free cash flow to the term loan until it is repaid. The notes are bullet, meaning they cannot be repaid early and are settled at maturity.

Q. Could a fund really write an US\$18.9 billion cheque?

Not alone. It would be a consortium. That changes who signs, not the arithmetic, and I say so in the model.

Part 4 · The three rules

These are what separate this repository from the hundreds of others with the same project names on them. Be able to say why each one matters.

1. Every number is a live formula, and a checks sheet proves it

No answer is typed in. Every model has a Checks sheet that has to read MODEL OK, testing things like: does the balance sheet balance, does enterprise value equal the sum of its parts, does the money multiple agree with the IRR, is every forecast cash flow positive.

Some checks are not arithmetic at all — they are discipline. Project 5 fails if forecast revenue growth ever exceeds anything TCS achieved in the last three years. Project 7 fails if the exit multiple is set above entry. Those exist because quietly lifting an assumption until the answer looks right is the most common way a model goes wrong, and a check is cheaper than a reviewer.

2. Every source is named, and the weak ones are flagged

Where a number is approximate, it says so. Project 4 marks which target revenues came from rounded press releases. Project 6 flags that the credit mark is an assumption on an estimated loan book and the largest judgement in the model. Project 5 states that no broker-target bar could be verified from a free source.

Admitting the weak spots before someone finds them is what makes the strong parts believable.

3. Every project says what it cannot do

Each notes page has a section on what it would take to do the analysis properly, and what the analysis does not support. Project 4 says its TCS range is illustrative because no target in the set is remotely TCS's size. Project 5 says 70% of the DCF is terminal value. Project 7 says the leverage sensitivity flatters high-leverage cases and explains why.

The point of all three

An interviewer is not checking whether your model is right. They are checking whether you know where it is wrong. Every senior banker has seen a confident model blow up. What they are hiring for is judgement about limits, and this portfolio is built to demonstrate exactly that.

Before any interview

- Open the model you are about to discuss and click through the Checks sheet. Know what each check tests.
- Know your three headline numbers cold: WACC, the valuation conclusion, and the one assumption the answer is most sensitive to.
- Be ready to say what you would do differently with more time. It is the question most candidates fumble and it is written on every notes page.
- Never claim a number you cannot source. 'I would have to check that' costs you nothing; a wrong figure stated confidently costs you the interview.